SKILL 23:

```
  GNU nano 8.7.1                skill-23.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
int pipe1[2];
int pipe2[2];
pipe(pipe1);
pipe(pipe2);

pid_t p1 = fork();
if (p1 == 0) {
dup2(pipe1[1], STDOUT_FILENO);
close(pipe1[0]);
close(pipe1[1]);
close(pipe2[0]);
close(pipe2[1]);
execlp("ls", "ls", NULL);
perror("ls");
exit(1);
}

pid_t p2 = fork();
if (p2 == 0) {
dup2(pipe1[0], STDIN_FILENO);
dup2(pipe2[1], STDOUT_FILENO);
close(pipe1[0]);
close(pipe1[1]);
close(pipe2[0]);
close(pipe2[1]);
execlp("grep", "grep", ".c", NULL);
perror("grep");
exit(1);
}

pid_t p3 = fork();
```

```
  GNU nano 8.7.1                skill-23.c
dup2(pipe1[0], STDIN_FILENO);
dup2(pipe2[1], STDOUT_FILENO);
close(pipe1[0]);
close(pipe1[1]);
close(pipe2[0]);
close(pipe2[1]);
execlp("grep", "grep", ".c", NULL);
perror("grep");
exit(1);
}

pid_t p3 = fork();
if (p3 == 0) {
dup2(pipe2[0], STDIN_FILENO);
close(pipe1[0]);
close(pipe1[1]);
close(pipe2[0]);
close(pipe2[1]);
execlp("wc", "wc", "-l", NULL);
perror("wc");
exit(1);
}

close(pipe1[0]);
close(pipe1[1]);
close(pipe2[0]);
close(pipe2[1]);

waitpid(p1, NULL, 0);
waitpid(p2, NULL, 0);
waitpid(p3, NULL, 0);

printf("Pipeline completed successfully.\n");
return 0;
}
```


harshitha18@LAPTOP-6E22AR6C:~$ nano skill-23.c
harshitha18@LAPTOP-6E22AR6C:~$ gcc skill-23.c -o skill-23
harshitha18@LAPTOP-6E22AR6C:~$ ./skill-23
87
Pipeline completed successfully.
harshitha18@LAPTOP-6E22AR6C:~$